

Lecture 12

Different Addressing / Targeting Modes:

This should explain about when a packet is sent over the internet who is meant to receive it.

Broadcast: every device on the local network gets the packet. Note it only works for devices on the same LAN. It works one to many

Multicast: one device sends packet/s to a defined group of devices, usually on the same network but it can be routed to other networks. Devices must join the multicast group

Unicast: it the most common type of casting. I device sends a data frame or a packet to another single device on the same LAN or on other network

Anycast: a device sends a packet to a group of devices that have the same ip, and only one receives it, the one with the shortest route.

IPv4 Addresses:

IPv4 addresses consists of 4 bytes (32 bits) with each byte separated by a dot when represented as a decimal number. Each byte can be a value between 0 and 255.

The IPv4 address consists of 2 parts the network ID and the host ID. How these parts are split is decided by the subnet masking. Which is split into 5 classes A,B,C,D,E

Network ID: this part of the IPv4 is the network identifier and is shared by all the devices on the same network.

Host ID: this is the last part of the IPv4 and it defines the host/device exactly on the network.

Now for the masking.

Subnet masking is usually represented as such 255.255.255.0

The byte that is represented by all ones is reserved for the network id and the one that is represented by the 0 is reserved for the host id.

So if we see something like this: 255.255.255.0 /24 this means the first 24 bits/ 3 bytes are reserved for the network id.

Classes in IPv4:

Classes were originally used to define network size by fixing how much of the IP address was the network and how much was the host, without subnetting.

Class A: it used for networks with many devices for very large networks it uses the first 7 bits which means it has the range of 0.0.0.0 → 127.255.255.255 and the mask of 255.0.0.0 / 8

Giving us the chance of including as many users as 2^{24} devices

And 2^7 Networks

We have 2 reserved addresses one for the network where the last 3 bytes are zeros and one as a broadcast where the last 3 bytes are 255

Class B: this is used for middle sized networks compared to the class A. it uses the first 2 bytes for the network Id and uses 14 bits to represent the network. Having a range from 128.0.0.0 → 239.255.255.255. it has the network masking of value 255.255.0.0 / 16

It can offer us a network of size 2^{14}

And devices up to 2^{16}

We have 2 reserved addresses one for the network where the last 2 bytes are zeros and one as a broadcast where the last 2 bytes are 255

Class C: this is used for smaller networks and it uses the first 3 bytes to represent the network id and the last one byte to represent the host id. It has network masking of 255.255.255.0 /24. It uses 21 bits to represent the network and 8 bits to represent the host

The address range it offers 192.0.0.0 – 223.255.255.255

We have 2 reserved addresses one for the network where the last 1 bytes are zeros and one as a broadcast where the last 1 bytes are 255

It offers 2^{21} addresses for the network

It offers 2^8 addresses for the host.

Class D: it is used for the purpose of multicasting where the first 4 bits are 1110.

Address range 224.0.0.0 – 239.255.255.255

Class E: it uses the first 4 bits as 1111. And are reserved for experimental purposes and future use. They are not meant for normal host assignment and are generally not routable on the internet, meaning routers will ignore them.

Address range 240.0.0.0 – 255.255.255.255

IPv4 Network and Broadcast Address:

Not every IP address on a network is available for a host some are for broadcasting some defines the network.

Broadcast and network examples from class C: 192.2.8.10

Here we cannot use an address for a host like this 192.2.8.0 cause this is reserved for the network.

We cannot use 192.2.8.255 cause this is reserved to broadcast a message on this local network so this means our host address range for this C network is 192.2.8.1 → 192.2.8.254

Some reserved addresses that are not for public use.

0.0.0.0/8 → represent this network. The first byte stays as zero

10.0.0.0/8 — **Private networks** range of 10.0.0.1 → 10.255.255.254 this is used for local networks that does not want to do any communication with outside networks.

100.64.0.0/10 — **Carrier Grade NAT (CGN)** → range of 100.64.0.0 – 100.127.255.255 . its used for ISPs.

127.0.0.0/8 — **Loopback addresses** → Range: 127.0.0.0 – 127.255.255.255. used by users to test some network services.

169.254.0.0/16 — **Automatic private IP (APIPA / Zeroconf)** → Range: 169.254.0.0 – 169.254.255.255 .

172.16.0.0/12 — **Private networks (RFC 1918)** → Range: 172.16.0.0 – 172.31.255.255. Purpose: Private IP addresses for internal LANs

192.0.0.0/24 — **On request for assignment process (RFC 5736)** → Range: 192.0.0.0 – 192.0.0.255. Purpose: Reserved for special assignments, typically for protocols or network management

192.168.0.0/16 — **Private networks (RFC 1918)** → Range: 192.168.0.0 – 192.168.255.255 .Purpose: Most common home/small office private network addresses

192.18.0.0/15 — **For testing network components** → Range: 192.18.0.0 – 192.19.255.255

Purpose: Reserved for testing network devices and protocols

224.0.0.0/4 — **Multicast addresses (RFC 3171)** → Range: 224.0.0.0 – 239.255.255.255 Purpose: Multicast addresses for one-to-many communication

240.0.0.0/4 — Reserved for future use (RFC 1112)→ Range: 240.0.0.0 – 255.255.255.254

Purpose: Reserved / experimental; Class E addresses

255.255.255.255/32 — Limited broadcast (RFC 1700)→ Single address: 255.255.255.255

Purpose: send a packet to all devices in the current network only

IPv4 Class System Disadvantages:

IPv4 address exhaustion in general.

Less flexible system:

Fixed subnet masks for each class; means that each network is going to reserve an exact number of hosts whether they need it or not based on the class they choose. For instance a class B network would give a company 2^{16} hosts even if they need only 2^{10} which means the difference is wasted.

Only 128 networks for class A; this means only 128 entity can have a network with class A which is not enough for what we have right now.

Class C is too small and class B is too big.

Hard to change network structure if needed; Because of fixed classes, splitting or resizing networks is difficult

Network migration is complicated; upgrading from network C to Network B means changing the DNS and each IP address that was before on the network.

IPv4 address exhaustion in general; with all the wasted ip addresses and with offering only 4.3 billion IP addresses it did not take very long for this version to be exhausted.

IPv4 Possible (intermediate) solutions:

Use of private networks and address translation; To save IPv4 addresses, organizations don't need globally unique public IPs for every device. Instead, they use private IP addresses inside their network.

New address system with →increased address space →Meeting future requirements. Referring to the IPv6 introduction.

Classless Interdomain Routing (CIDR):

allows logical networks to be divided into subnets, creating a flexible hierarchy of network → subnet → host, while still retaining the original network ID. Unlike the rigid classful system (A, B, C), CIDR uses a subnet mask to split the host portion of an IP address into a subnet ID and a host ID, enabling more efficient use of addresses. The size of the network portion (network + subnet) is indicated using a slash notation (e.g., /24), which specifies how many bits are dedicated to the network, while the remaining bits represent hosts within each subnet.

When creating subnets the difference between what the original bits for a class is and what you borrowed power 2 is how many subnets are there.

Example: Class B network subnet mask 21 bit = 16 bit + 5 bit

5 bit → $2^5 = 32$ subnets

11 bit for host id (from original Host ID) → $2^{11} = 2048$ host addresses per subnet

Address Resolution with ARP:

The Address Resolution Protocol (ARP) is used in IPv4 to map a device's network-layer IP address to its data-link-layer MAC address, allowing devices on the same local network to communicate. Essentially, ARP connects the network layer (IP) with the data link layer (Ethernet). In IPv6, this function is handled by the Neighbor Discovery Protocol (NDP), which replaces ARP and provides additional features such as address autoconfiguration and neighbor reachability detection.

Diagnostics in Network with ICMP Layer 3:

The Internet Control Message Protocol (ICMP) is a tool used in networks to help maintain and troubleshoot connectivity. It can detect transmission errors, provide basic diagnostic information, and let devices report problems to each other. Common uses include ping (checking if a device is reachable) and traceroute (seeing the path packets take), but the deeper technical details of ICMP are beyond the scope of this lecture.

Addressing with IPv6:

It uses 16 bytes to represent an address. 128 bits. Unlike the IPv4 it represents addresses in hexadecimal. To shorten it we can eliminate any zeros from the left from any section of the 8 used to represent it. If there is a series of 0s we just make it empty :: if we have 2 series of zeros we eliminate the longest only the other one can be represented as :0:

Final 4 bytes might be in decimal (to embed IPv4).

Structure of IPv6 Address and Networks I:

The default format for IPv6 is 64-64 or 8-8 bits/bytes

Where the first 8 bytes are reserved to represent the network, and the last 8 bytes are used to represent the host. This is the default representation but it's not a solid rule and can be changed/costumised

IPv6 was not used to fully replace IPv4 but to run in parallel with it. Most devices and network model can run both of them.

IPv6 uses CIDR-style addressing like IPv4: flexible prefixes rather than rigid classes. You can assign subnets of any size using slash notation, e.g., /32, /48, /64.

End user usually gets a /56 or /64 network assigned: When your ISP gives you IPv6: /64 → small network (single LAN) → 2^{64} possible addresses for your devices /56 → larger network → lets you create multiple /64 subnets in your home or small office.

The same as in IPv4 CIDR the difference between a class network bits and what you costumise is more network subnets

Scopes of IPv6 Addresses I:

Definition of a topological area, where an address is valid

Global Unicast Address (GUA): it public uni-cast normal IP addresses which can send and receive public addresses on the internet. typical 2000::/3

Unique Local Address (ULA): private IP addresses it not publicly reachable and only used in LANs mostly in organizations. Prefix: fc00::/7

Link-Local Address (LLA): it's a special mandatory local address that exists on every interface even if you already have a ULA or GUA. Prefix: fe80:/10 . No forwarding of packets by router of these addresses

Host Address: The loopback address is used inside a device to refer to itself. It's similar to IPv4's 127.0.0.1. Every device's network stack uses it to test itself without sending packets over the network. ::1/128

Multicast Addresses: IPv6 does not have broadcast like IPv4. Multicast replaces broadcast by sending a message to a specific group of devices instead of all devices. Multicast prefix ff::/8

Flag Bits

- Usually 0000

- Defines special options or behavior (not critical for basic understanding).

Scope Bits

- Defines how far the multicast message can travel:
 - 0001 → interface-local → stays inside the device (like loopback for multicast)
 - 0010 → link-local → stays inside the local network segment; routers do not forward these packets
- Other scope bits exist for larger networks (site-local, organization-local, global), but interface-local and link-local are most basic.

Network Address Translation (NAT):

Scenario:

- Households use a private IPv4 address space (private network)
- Communication of network devices with devices which have a globally accessible address.

Network Address Translation:

- Router is equipped with public IP-Address, typically by Internet Service Providers (ISP)
- Router is default gateway private network.

that means is when i used my private IP address to try to communicate with a server outside my LAN my router which is the gateway to the public internet replaces my private IP address with a public ip address then send my request to the public internet and when my request comes back with a reply from the public internet the Router uses the NAT table to revert this process and then forward the reply using my private IP.

the device also gets a port number in the NAT

Port and address is exchanged

- Advantage: private network topology + number of devices is not visible to the public network